NATIONAL WORKSHOP

PROBLEM STATEMENTS & PROPOSED SOLUTIONS

Mining Industry — Research Questions & Solution Approaches

Date: 23 June 2026

1. Prakriti Singh (DMT)

S.No.	Research Question	Proposed Solution
1	*How can women-specific safety frameworks be developed to improve inclusivity and security in mining operations?*	Develop gender-responsive safety protocols including dedicated facilities, harassment prevention mechanisms, and women-led safety committees. Conduct regular audits to benchmark inclusivity metrics and update policies accordingly.
2	*What strategies can mitigate occupational health issues among mining workers in remote locations?*	Deploy mobile health clinics, telemedicine platforms, and periodic health camps at remote mine sites. Establish mandatory health monitoring programs and tie worker welfare indices to operational performance reviews.
3	*How can mining companies optimize operating costs while maintaining safety and productivity?*	Implement lean mining principles, energy-efficient equipment, and real-time cost-tracking dashboards. Integrate predictive analytics to eliminate unplanned downtime and reduce idle equipment hours.
4	*What advanced technologies can minimize the environmental impacts of mining activities?*	Adopt IoT-based environmental monitoring systems, drone-assisted reclamation planning, and AI-driven emission control. Integrate green mining standards into mine planning from the design stage.
5	*How can Resettlement and Rehabilitation (R&R) programs be made more sustainable and socially acceptable?*	Co-design R&R plans with affected communities using participatory frameworks, ensuring livelihood restoration and transparent grievance mechanisms. Align programs with UN Guiding Principles on Business and Human Rights.
6	*What policy reforms can streamline mining license approvals and reduce regulatory delays?*	Introduce single-window clearance portals, fixed-timeline regulatory commitments, and digital document management for license applications. Establish inter-

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		departmental coordination cells to resolve bottlenecks proactively.
7	*How can proactive hazard identification systems reduce workplace accidents in mines?*	Deploy AI-powered hazard mapping tools, wearable sensor networks, and near-miss reporting apps integrated with real-time dashboards. Conduct structured HAZOP studies at each operational phase.
8	*What adaptive mine planning techniques can address weather-related operational disruptions?*	Integrate real-time meteorological data feeds into mine scheduling software and develop contingency operational plans for monsoon and extreme weather events. Design drainage and haul-road systems for climate resilience.
9	*How can predictive maintenance improve equipment reliability and operational efficiency?*	Deploy IoT sensors and machine-learning algorithms to monitor equipment health parameters and predict failures before breakdown. Establish centralized predictive maintenance control rooms linked to OEM diagnostic platforms.
10	*What workforce development models can enhance employee skills and accountability in mining operations?*	Implement competency-based training frameworks, digital simulation-based learning, and performance-linked incentive structures. Establish internal certification programs aligned with national mining skill standards.
11	*How can geological uncertainty and seismic risks be incorporated into mine planning frameworks?*	Utilize 3D geological modelling, seismic risk zonation maps, and probabilistic risk assessment tools in mine design. Build flexibility buffers into production schedules to accommodate geological variability.

2. Rahul Kumar

S.No.	Research Question	Proposed Solution
1	*How can integrated mine-water management systems reduce water seepage and improve water utilization?*	Design closed-loop water circuits with sump monitoring, hydrogeological modelling, and automated pump controls. Treat and recycle mine water for dust suppression and industrial use to minimize freshwater dependency.
2	*What innovative dust suppression technologies are effective under extreme temperature conditions?*	Deploy chemical foam suppressants, automated misting cannons, and moisture-retaining road binders suited to high-temperature environments. Integrate dust monitoring sensors with suppression systems for automated response.
3	*How can blast design be optimized to minimize boulder generation and improve loading efficiency?*	Use precision delay detonators, electronic blast timing systems, and 3D fragmentation modelling software to optimize burden-spacing ratios. Conduct post-blast fragmentation analysis to iteratively refine blast parameters.
4	*What sustainable alternatives exist for external dump management in land-constrained mining operations?*	Adopt concurrent backfilling strategies, paste technology for tailings management, and progressive dump reclamation to reduce land footprint. Explore geotechnically engineered inner-dump configurations in mined-out areas.
5	*How can intelligent traffic management systems reduce LMV-related risks inside mines?*	Install GPS-based fleet tracking, speed-limit enforcement sensors, and AI-powered collision avoidance systems for light mining vehicles. Create segregated traffic corridors and mandate proximity detection devices on all vehicles.

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6	*What blast optimization techniques can prevent back-break and bench instability?*	Apply controlled blasting methods such as line drilling and pre-splitting, combined with geomechanical assessment of bench conditions. Monitor vibration levels and adjust powder factor to protect slope integrity.
7	*How can predictive diagnostics improve the reliability of electronic safety devices in dumpers?*	Integrate onboard telematics with OEM diagnostic tools for real-time health monitoring of safety-critical components. Establish predictive replacement schedules based on usage cycles and sensor-triggered alerts.
8	*What workforce management strategies can improve the productivity of contractual mining workers?*	Implement performance-based contracts with clear KPIs, structured induction training, and regular supervisory reviews. Align contractor welfare standards with permanent employee norms to improve retention and output quality.
9	*How can regulatory processes related to forest clearance and land acquisition be accelerated?*	Engage regulatory bodies early through pre-application consultations, use digital submission portals, and appoint dedicated compliance liaisons. Build strong environmental baseline data packages to reduce approval cycle times.
10	*What is the carbon reduction potential of electrified mining fleets compared to diesel-operated equipment?*	Electrified fleets can reduce Scope 1 emissions by 30–60% depending on grid mix, with further reductions possible through renewable energy integration. Lifecycle cost analyses consistently show long-term total cost advantages for electrification.
11	*How does uncontrolled dust affect worker productivity and mine safety performance?*	Uncontrolled dust reduces visibility, impairs respiratory health, and increases accident risk, collectively reducing worker productivity by up to 20%. Systematic dust control programs directly improve safety KPIs and output consistency.
12	*How can misfire detection and management systems improve blasting safety?*	Deploy electronic detonator systems with built-in misfire diagnostics and establish standardized misfire reporting and neutralization protocols. Train blasting crews in systematic dud-hole inspection procedures after every blast.
13	*What mine-planning approaches can reduce haulage distances caused by adverse geological conditions?*	Adopt 3D geological models to dynamically re-route haul roads, optimize pit geometry, and minimize waste-to-ore haul distances. Apply fleet dispatch simulation tools to evaluate alternative haul configurations under varied geological scenarios.

3. Pankaj Jaiswal

S.No.	Research Question	Proposed Solution
1	*How can advanced safety management systems improve HEMM operational safety?*	Implement integrated safety management platforms with real-time HEMM tracking, operator fatigue monitoring, and automated safety interlocks. Conduct structured HEMM safety audits tied to operator certification renewals.
2	*What innovative approaches can address land acquisition challenges in mining projects?*	Adopt community-inclusive land acquisition frameworks with fair compensation benchmarks, transparent grievance systems, and phased acquisition timelines. Engage local panchayats and community representatives as co-facilitators in the acquisition process.

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3	*How can behavioral safety programs reduce accident frequency in mines?*	Deploy behavior-based safety (BBS) observation programs, near-miss reporting culture initiatives, and peer-led safety mentoring. Use data analytics to identify behavioral risk patterns and tailor targeted interventions.
4	*What methods can improve explosive efficiency while minimizing blasting hazards?*	Utilize emulsion explosives with optimized energy release profiles, combined with electronic delay systems for precise detonation control. Conduct post-blast performance assessments to continuously calibrate explosive charge designs.
5	*How can environmental monitoring systems support sustainable mining practices?*	Install real-time multi-parameter environmental monitoring networks covering air, water, noise, and soil parameters with automated regulatory reporting. Link monitoring data to operational triggers to enable immediate corrective responses.
6	*What optimization techniques can improve underground mine productivity?*	Apply stope design optimization, autonomous equipment integration, and real-time production tracking systems to maximize ore extraction efficiency. Implement digital twin models of underground workings for scenario-based planning.
7	*How can digital technologies transform operational decision-making in mines?*	Integrate ERP systems, AI-driven analytics dashboards, and drone-based surveying to enable data-driven operational decisions. Establish a unified digital operations center to consolidate all mine data streams in real time.
8	*What opportunities exist for indigenous manufacturing of HEMM equipment in India?*	Leverage the PLI scheme and MAKE IN INDIA policy framework to incentivize domestic HEMM component manufacturing and assembly. Establish industry-academia R&D partnerships to develop indigenous heavy equipment prototypes suited to Indian mining conditions.
9	*What is the feasibility of mine electrification for reducing carbon emissions?*	Mine electrification is highly feasible for fixed infrastructure and surface tramways, with battery-electric and trolley-assist systems offering viable pathways for mobile HEMM electrification. Pilot programs at Indian opencast mines should be supported with government co-funding.
10	*How can skill development programs bridge workforce competency gaps in mining?*	Establish competency mapping frameworks aligned with NCVT standards, supported by simulation-based training centers and RPL certification for experienced workers. Partner with ITIs and mining institutes for structured apprenticeship pipelines.
11	*What technologies can enhance methane exploration and utilization in coal mines?*	Deploy advanced borehole seismic and geophysical surveys for methane delineation, combined with in-seam gas drainage systems for pre-mining extraction. Utilize captured methane for on-site power generation or supply to regional gas grids.
12	*How can abandoned mine lands be sustainably reclaimed and reused?*	Develop mine closure plans that integrate land reclamation with ecological restoration, solar farm development, and community livelihood programs. Apply successful international frameworks such as Germany's Lusatia model for post-mining land reuse.

4. Abisri R. (DMT Consulting Pvt. Ltd.)

S.No.	Research Question	Proposed Solution
1	*How can real-time slope stability monitoring reduce dump failure risks in opencast mines?*	Deploy radar-based slope monitoring systems, IoT tiltmeters, and piezometric sensors with automated alarm thresholds. Integrate monitoring data with geotechnical models for early warning and emergency response activation.
2	*What integrated dewatering strategies can effectively manage aquifer-related challenges?*	Combine hydrogeological modelling, multi-level pumping systems, and groundwater recharge interventions to manage aquifer drawdown. Establish water balances and adapt dewatering schedules based on real-time groundwater level data.
3	*How can dust, noise, and air pollution be minimized through advanced environmental technologies?*	Implement automated dust suppression systems, acoustic barriers, and low-emission equipment fleets integrated with real-time environmental monitoring. Apply predictive pollution modelling to pre-emptively schedule high-impact operations during favorable atmospheric conditions.
4	*What occupational health frameworks can reduce workplace health risks in mining operations?*	Establish routine health surveillance programs, occupational exposure limit monitoring, and personal protective equipment compliance systems. Integrate occupational health data with operational records to identify and mitigate hazard-health linkages.
5	*How can backfilling and reclamation practices support sustainable mine closure?*	Design phased progressive backfilling programs using mine waste and overburden, combined with topsoil replacement and native species revegetation. Align mine closure plans with post-mining land use agreements with local communities and regulators.
6	*What innovative support and ventilation systems can improve underground mine safety?*	Adopt primary and auxiliary ventilation optimization software, digital airflow monitoring sensors, and advanced ground support systems such as fiber-reinforced shotcrete and resin-grouted rockbolts. Simulate ventilation scenarios using CFD models.
7	*How can mining approval and permitting processes be optimized without compromising compliance?*	Digitize permitting workflows, establish pre-application regulatory guidance cells, and create standardized compliance checklists to reduce approval timelines. Introduce performance-based permitting that rewards compliant operators with expedited renewals.
8	*What methods can reduce ore dilution and grade fluctuations in underground mines?*	Implement selective mining methods with precise blast confinement, real-time grade control using XRF analyzers, and ore tracking from stope to mill. Use 3D geological models to refine ore boundary delineation and reduce contact dilution.
9	*How can blast design optimization improve fragmentation while reducing vibration impacts?*	Apply electronic detonator systems with millisecond-precision delay sequencing and use vibration prediction models to design blasts within allowable vibration limits. Conduct post-blast vibration monitoring and fragmentation analysis to validate and refine designs.
10	*What training frameworks can enhance HEMM operator competency?*	Develop structured competency assessment matrices, simulator-based training programs, and periodic on-machine performance evaluations. Link operator certification to continuous professional development requirements with performance score thresholds.
11	*How can mine subsidence risks be predicted and mitigated?*	Use subsidence prediction models, InSAR satellite monitoring, and time-domain ground movement sensors to detect and quantify subsidence. Design mine layouts

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		with protective pillars and backfilling strategies to minimize surface settlement.
12	*What monitoring technologies can prevent methane accumulation in underground workings?*	Install distributed methane sensor networks with real-time telemetry, automatic ventilation boost systems, and gas drainage pre-drainage boreholes. Establish strict methane threshold-based stoppage protocols integrated with mine control systems.

5. Suraj Subhash Dhanbhoj

S.No.	Research Question	Proposed Solution
1	*What innovative approaches can overcome land acquisition challenges in mining projects?*	Develop community partnership models, transparent compensation frameworks, and early stakeholder engagement strategies. Use digital grievance management systems to build trust and reduce resistance during acquisition processes.
2	*How can statutory clearance processes be streamlined to reduce project delays?*	Adopt integrated digital clearance portals, inter-ministry coordination protocols, and time-bound statutory review mechanisms. Appoint dedicated project clearance officers to manage complex multi-clearance submissions.
3	*What blast optimization techniques can minimize boulder generation and excavation delays?*	Apply stemming optimization, controlled burden-spacing ratios, and electronic delay sequencing to achieve consistent fragmentation. Use digital photogrammetry for pre-blast geological mapping to tailor blast designs to rock conditions.
4	*How can dust generation be effectively controlled in large-scale surface mines?*	Implement integrated dust management plans combining road watering schedules, chemical binders, water cannons, and enclosure of transfer points. Operate real-time dust monitoring networks linked to automated suppression systems.
5	*What strategies can address habitat and biodiversity conservation challenges in mining regions?*	Conduct biodiversity baseline studies, establish buffer zones and wildlife corridors, and implement progressive reclamation with native species. Engage ecologists in mine planning to ensure compliance with environmental impact assessment requirements.
6	*How can mine-water seepage be effectively managed and recycled?*	Construct lined sump systems, underground seepage barriers, and water treatment plants to capture and recycle mine water. Integrate mine water balances with operational scheduling to maximize reuse and minimize discharge.
7	*What predictive maintenance techniques can improve haul-road performance during monsoons?*	Monitor haul-road moisture content and CBR values using embedded sensors and schedule pre-monsoon drainage improvement works using predictive weather models. Deploy rapid-repair protocols activated by road condition alerts during heavy rainfall periods.
8	*How can dump stability be improved considering varying shale and material properties?*	Conduct geotechnical characterization of dump materials, apply slope angle optimization based on shear strength parameters, and install real-time stability monitoring. Implement engineered toe berms, drainage trenches, and benched dump profiles to improve factor of safety.
9	*What intelligent transportation systems can enhance LMV road safety within mines?*	Install GPS fleet tracking, speed-monitoring cameras, and automated barriers at high-risk intersections within

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		mine roads. Enforce segregated corridors for light vehicles and heavy equipment with time-based access controls.
10	*How can mine illumination systems improve worker safety and productivity?*	Deploy high-lumen LED lighting across operational areas, haul roads, and maintenance zones with automatic dimming and fault detection. Conduct illumination mapping to identify and eliminate shadow zones that increase accident risk.
11	*What training methodologies are most effective for developing a skilled mining workforce?*	Combine simulator-based skill training, on-the-job mentoring, and competency assessments to build a multi-skilled workforce. Institutionalize continuous learning programs with digital learning management systems accessible to all mine employees.
12	*How can dispatch road maintenance improve fleet efficiency and reduce operating costs?*	Implement road condition monitoring systems, proactive grading schedules, and material optimization for road surfaces to reduce rolling resistance. Use fleet management data to prioritize maintenance of high-traffic haulage corridors.

6. Alam Md. (Reliance Power Limited)

S.No.	Research Question	Proposed Solution
1	*How can accident prediction models reduce mine fatalities and injuries?*	Develop machine-learning-based accident prediction models using historical incident data, near-miss reports, and operational parameters. Integrate predictions with real-time risk dashboards to enable proactive safety interventions.
2	*What strategies can address skilled manpower shortages in the mining industry?*	Establish industry-funded mining skill development centers, structured apprenticeship programs, and partnerships with technical institutes. Offer competitive career pathways and rural outreach programs to attract talent to mining careers.
3	*How can air pollution and dust emissions be effectively controlled in mining operations?*	Deploy automated dust suppression systems, enclose material transfer points, and adopt low-emission equipment with after-treatment systems. Implement a real-time air quality management plan with regulatory reporting integration.
4	*What integrated frameworks can improve mine management and land reclamation practices?*	Adopt Mine Closure Plans from project inception, integrating progressive reclamation milestones with operational phases. Link reclamation performance to regulatory compliance frameworks and community benefit sharing agreements.
5	*How can rising fuel and energy costs be reduced through operational optimization?*	Implement energy audits, shift scheduling optimization, electric-assist haul systems, and mine-wide energy management systems. Explore on-site renewable energy generation to offset diesel dependency and stabilize energy costs.
6	*What regulatory reforms can accelerate mining project approvals?*	Introduce time-bound approval mandates, single-window online portals, and fast-track clearance pathways for compliant operators. Digitize inter-agency data sharing to eliminate redundant submissions and reduce approval bottlenecks.

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7	*How can emerging technologies improve recovery and resource utilization?*	Apply sensor-based ore sorting, AI-driven grade control, and advanced comminution technologies to improve recovery rates. Leverage digital twins and process simulation to continuously optimize mineral processing performance.
8	*What approaches can ensure sustainable mining under increasingly stringent environmental regulations?*	Integrate environmental management systems (ISO 14001), real-time compliance monitoring, and stakeholder reporting into standard operating procedures. Proactively adopt voluntary sustainability frameworks such as IRMA to demonstrate responsible mining.
9	*How can commodity price volatility be managed through strategic mine planning?*	Develop flexible mine plans with multiple production scenarios, maintain low-cost reserve portfolios, and implement financial hedging strategies. Use commodity price forecasting tools to dynamically adjust production volumes and cost structures.
10	*What innovative methods can reduce ground-water depletion caused by mining activities?*	Implement artificial groundwater recharge schemes, lined tailings ponds, and closed-circuit water systems to minimize drawdown. Conduct regular hydrogeological monitoring and adjust mine dewatering plans to protect aquifer integrity.
11	*How can remote mining operations contribute to socio-economic development in nearby communities?*	Establish community development funds, local procurement preferences, and livelihood training programs linked to mine operations. Develop shared infrastructure such as roads, power, and water supply that also benefit surrounding communities.

7. Sangram Pradhan (OCPL)

S.No.	Research Question	Proposed Solution
1	*How can fire hazards associated with HEMM operations be prevented through advanced monitoring systems?*	Install automatic fire suppression systems on HEMM, thermal imaging cameras for hotspot detection, and fire risk monitoring platforms. Conduct periodic fire risk assessments and mandate fire suppression equipment maintenance schedules.
2	*What workforce management strategies can improve contractor compliance with safety standards?*	Integrate contractors into the mine's Safety Management System with mandatory safety inductions, performance monitoring, and contractual safety KPI requirements. Establish contractor safety forums for collaborative issue resolution.
3	*What advanced dust suppression technologies are most effective in surface mines?*	Deploy dry fog suppression systems, chemical stabilizers, and high-pressure water cannons at key dust generation points. Integrate automated suppression triggers linked to real-time PM10/PM2.5 monitoring data.
4	*How can safety-factor-based mine design improve operational safety?*	Apply quantitative risk assessment and stability modelling during mine design to ensure all structures meet target safety factors. Review and update safety factors periodically based on operational monitoring data and changing ground conditions.
5	*What training models can enhance competency among mining operators and supervisors?*	Implement tiered competency training programs with simulator-based assessments, on-site mentoring, and periodic refresher certifications. Develop leadership training for supervisors focused on safety culture, decision-making, and team management.

S.No.	Research Question	Proposed Solution
6	*How can LMV traffic management systems improve safety within mine premises?*	Deploy automated speed enforcement, GPS-based traffic monitoring, and intelligent signaling at high-risk intersections. Introduce designated pedestrian pathways and enforce strict LMV-HEMM segregation protocols.
7	*What drilling optimization techniques can enhance productivity while maintaining safety?*	Utilize down-the-hole (DTH) hammer optimization, automated drill monitoring systems, and GPS-guided drill positioning to maximize penetration rates and accuracy. Integrate drilling data with blast design software for continuous improvement.
8	*How can sensor-based safety systems improve hazard detection in mining equipment?*	Fit HEMM with proximity sensors, collision avoidance radars, and fatigue detection systems integrated with operator alert mechanisms. Link sensor data to a centralized mine safety monitoring platform for fleet-wide hazard visibility.
9	*What visibility-enhancement technologies can reduce accidents involving large dumpers?*	Install 360° camera systems, radar blind-spot detection, and enhanced LED spotlighting on large dumpers. Introduce reversing alert systems and physical berm barriers at high-risk reversing zones within mine operations.
10	*How can tyre monitoring systems improve safety and maintenance efficiency?*	Deploy real-time tyre pressure and temperature monitoring systems (TPMS) on all HEMM with automated alerts for deviations. Implement predictive tyre life management programs based on usage data, load cycles, and haul-road conditions.
11	*What role can 360° camera systems play in preventing HEMM-related accidents?*	360° camera systems provide operators with comprehensive situational awareness, eliminating blind spots and reducing collision risk with personnel and other equipment. Integrate camera feeds with AI-based object detection for real-time alert generation.

8. R. L. Sen

S.No.	Research Question	Proposed Solution
1	*How can scientific land reclamation improve post-mining land productivity?*	Develop soil fertility restoration plans using organic amendments, native species revegetation, and topsoil stockpiling for reuse. Monitor reclaimed land productivity through vegetation indices and soil quality benchmarks over a defined post-closure period.
2	*What technologies can enhance conveyor system efficiency in surface mines?*	Deploy variable frequency drives, intelligent belt alignment systems, and automated load optimization controls on conveyor systems. Implement predictive maintenance for conveyor components using vibration analysis and thermal imaging.
3	*How can groundwater depletion caused by mining activities be minimized?*	Implement comprehensive mine dewatering management plans incorporating groundwater recharge zones, lined sump construction, and treated water recycling. Conduct quarterly hydrogeological assessments to adapt pumping strategies to aquifer recovery status.
4	*What strategies can improve coal handling plant performance and reliability?*	Apply predictive maintenance protocols, automated process control systems, and regular performance benchmarking to maximize coal handling plant availability. Optimize crushing and screening circuits using process simulation software.

S.No.	Research Question	Proposed Solution
5	*How can waste management practices support sustainable mining operations?*	Adopt waste hierarchy principles — reduce, reuse, recycle — for mine overburden, reject coal, and process effluents. Develop waste-to-resource programs converting mining by-products into construction materials, backfill media, or energy inputs.
6	*What advanced transportation systems can optimize mine dispatch operations?*	Deploy AI-driven fleet dispatch systems with real-time traffic optimization, fuel consumption tracking, and dynamic route assignment. Integrate dispatch data with production planning tools to align truck cycles with shovel and crusher availability.